

- 3 The cilia of ciliated epithelial cells show variation in length, within an individual and between different individuals.

Samples of ciliated epithelial tissue were removed from the airways of healthy people and the mean cilia length for each individual was calculated.

The people in the study formed two groups:

- people who were exposed to a harmful environmental factor
- people who were **not** exposed to a harmful environmental factor.

The results are shown in Fig. 3.1.

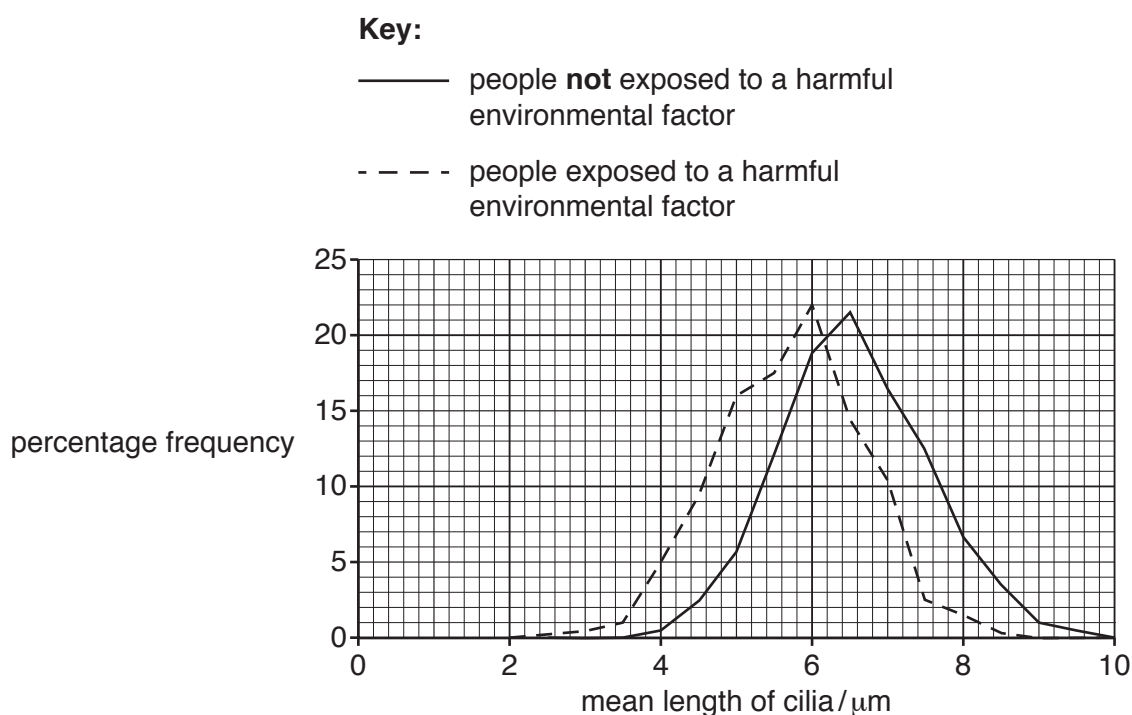


Fig. 3.1

- (a) (i) Name the type of variation shown for mean cilia length in each group of people in Fig. 3.1.

..... [1]

- (ii) Suggest **two** harmful environmental factors that people might have been exposed to that could explain the difference between the two curves in Fig. 3.1.

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..... [2]

- (b)** The development and final length of cilia of epithelial cells is controlled by many genes.

Fig. 3.2 summarises the interactions of some of these genes. The arrows represent the genes being switched on.

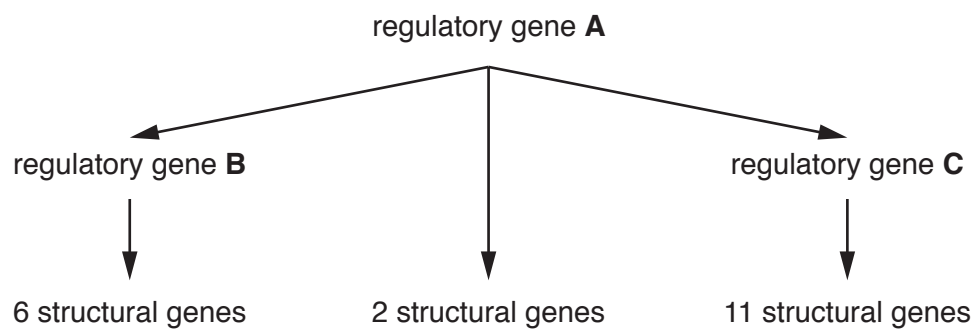


Fig. 3.2

- (i) With reference to Fig. 3.2, explain how genes such as **A**, **B** and **C** are able to switch on other genes.

[4]

- (ii) Describe how microarray analysis can be used to identify the genes switched on by the product of gene **C**.

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- (c) Microarray analysis has shown that ciliated epithelial cells and other epithelial cells in the airways express genes that code for receptor proteins on cell surface membranes. These receptor proteins detect specific chemicals.

When the receptor proteins bind to specific chemicals, this causes coughing, constriction of the airways and an increase in cilia activity. In people who have asthma, the response can be severe and life-threatening.

Suggest how knowledge of the genes for these receptor proteins could help in the treatment of asthma.

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[Total: 15]